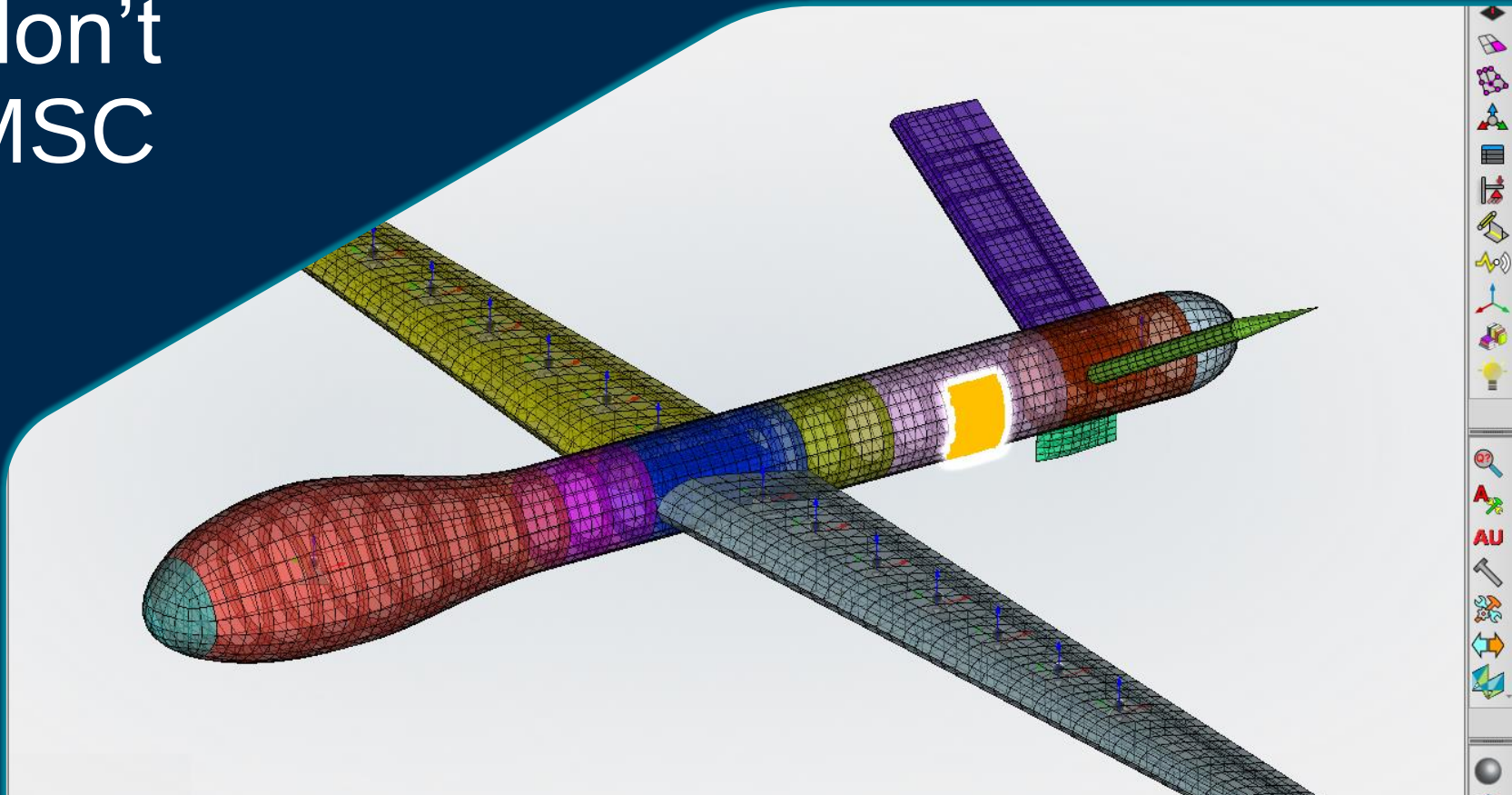


Five things you don't want to miss in MSC Nastran

Vasiliki Tsianika



Agenda

- 01 Single-Step TPA

- 02 Global-Local Analysis

- 03 Real Coupled Modes

- 04 HDF5 Output Modularity

- 05 ODS and PEAKOUT automation with ACMS

Single-Step TPA

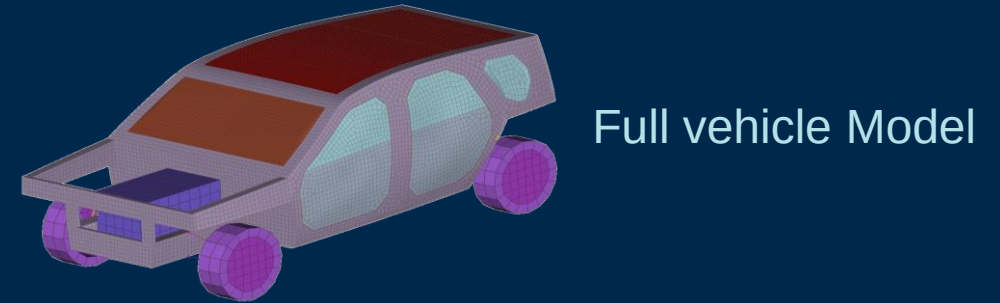
Transfer Path Analysis (TPA)

Transfer Path Analysis (TPA) is a method that separates a noise or vibration problem into its main contributors. By using source loads and transfer functions, it identifies which factors have the greatest impact on the response at a specific target location.



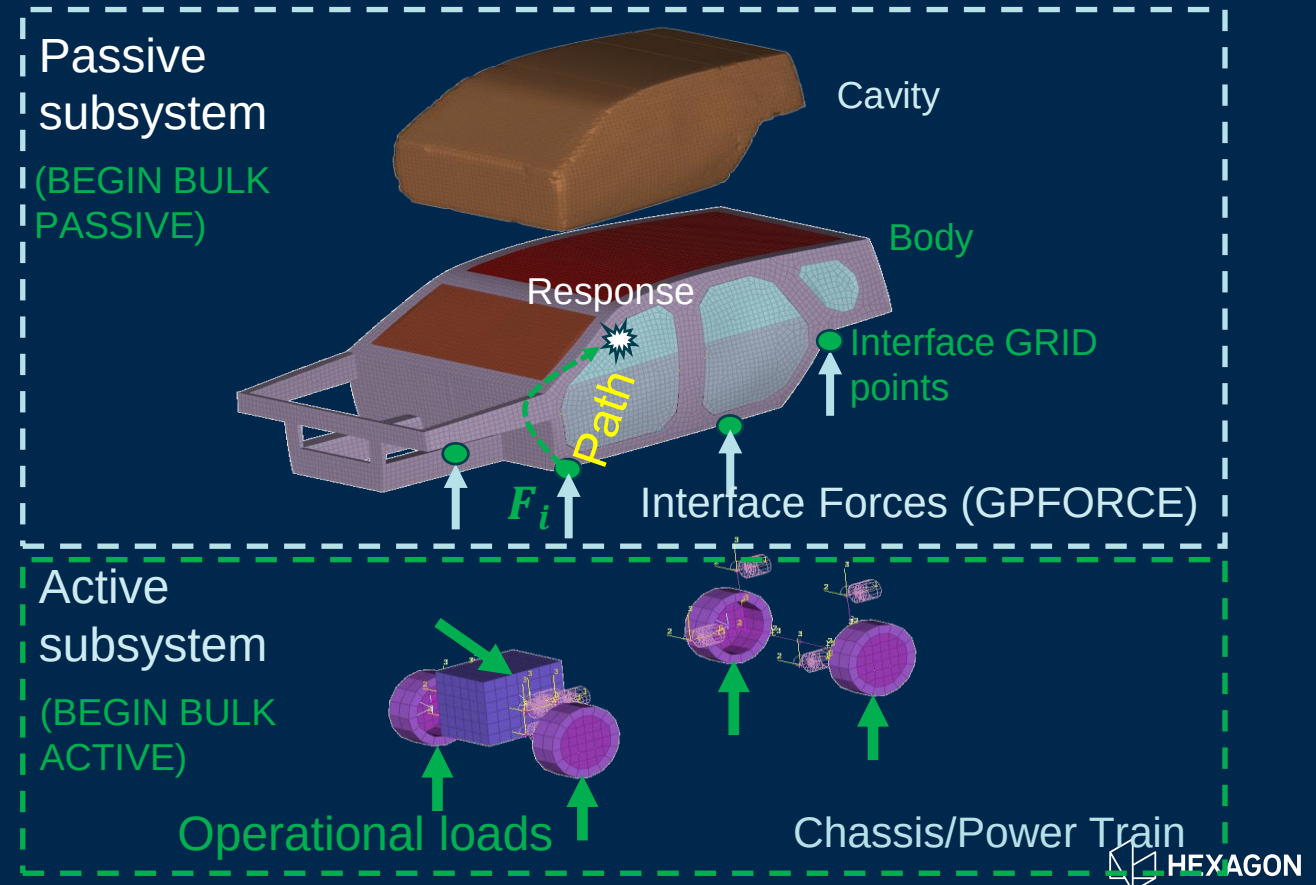
- ➡ In typical automotive applications, TPA analysis requires two separate frequency response SOL 111 runs
- ➡ MSC Nastran combines the individual steps into a single one called **Single-Step TPA**

Model Layout for Single-Step TPA

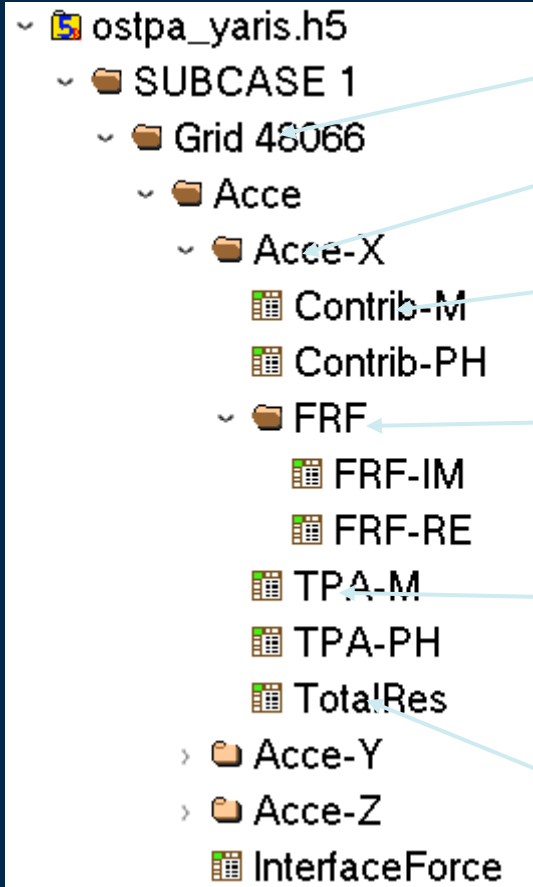


Full vehicle model consists of **four main sections**:

1. Case Control for a SOL 111 run:
 - New **TPASET** command
2. Main Bulk Data Section
 - New **TPADEFN** entry
 - HDF5OUT INPUT YES
3. **BEGIN BULK ACTIVE** (contains active model, including connector elements and interface grid points to the active model)
4. **BEGIN BULK PASSIVE** (contains passive model including user-defined interface grid points)



TPA results in the hdf5 output file



Grid point ID of response

Response Type and Direction

Path contributions (%) at the peak responses, in Magnitude and Phase

Frequency response functions due to unit load at paths

Path contributions (%), in Magnitude and Phase

Total response from the sum of individual path contributions

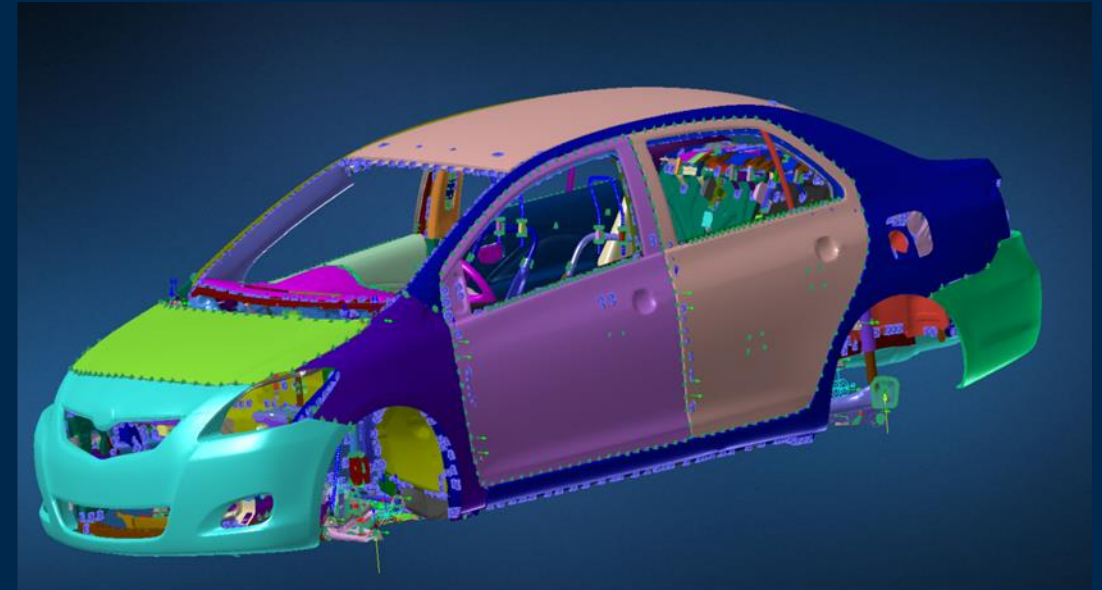
Example of Single-Step TPA

Model size:

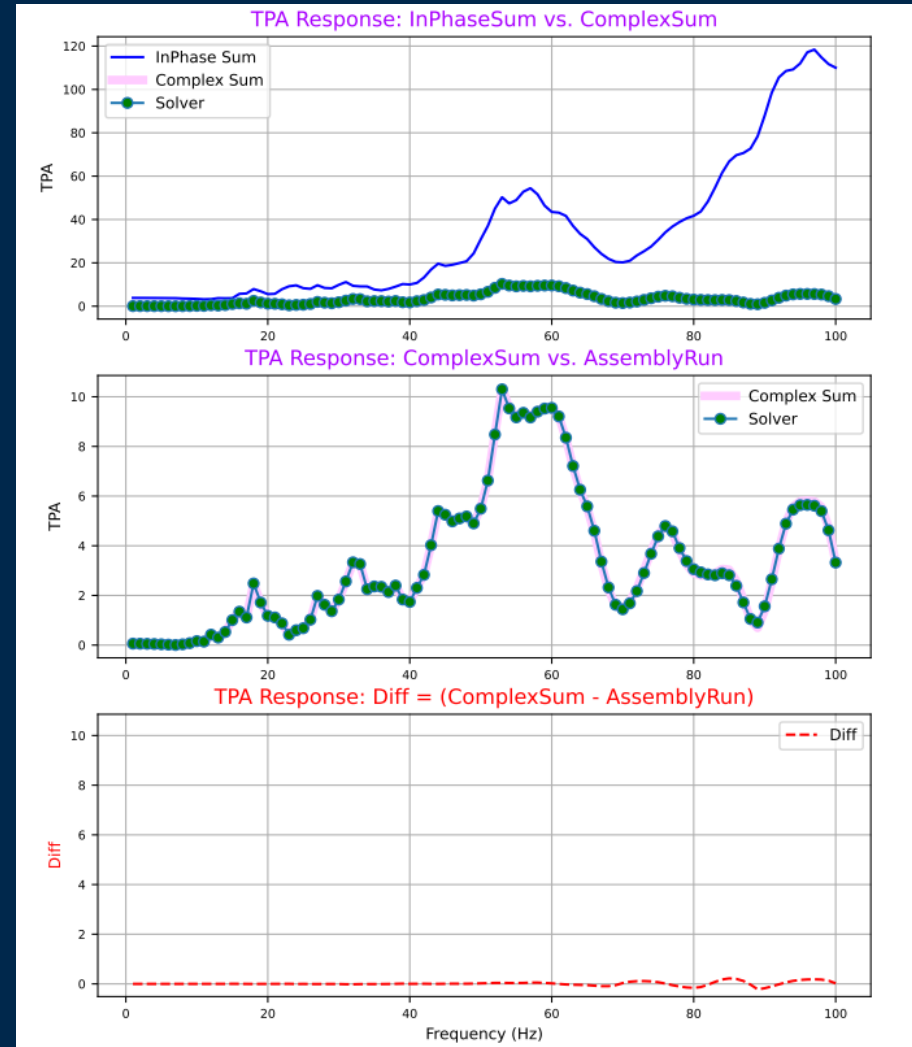
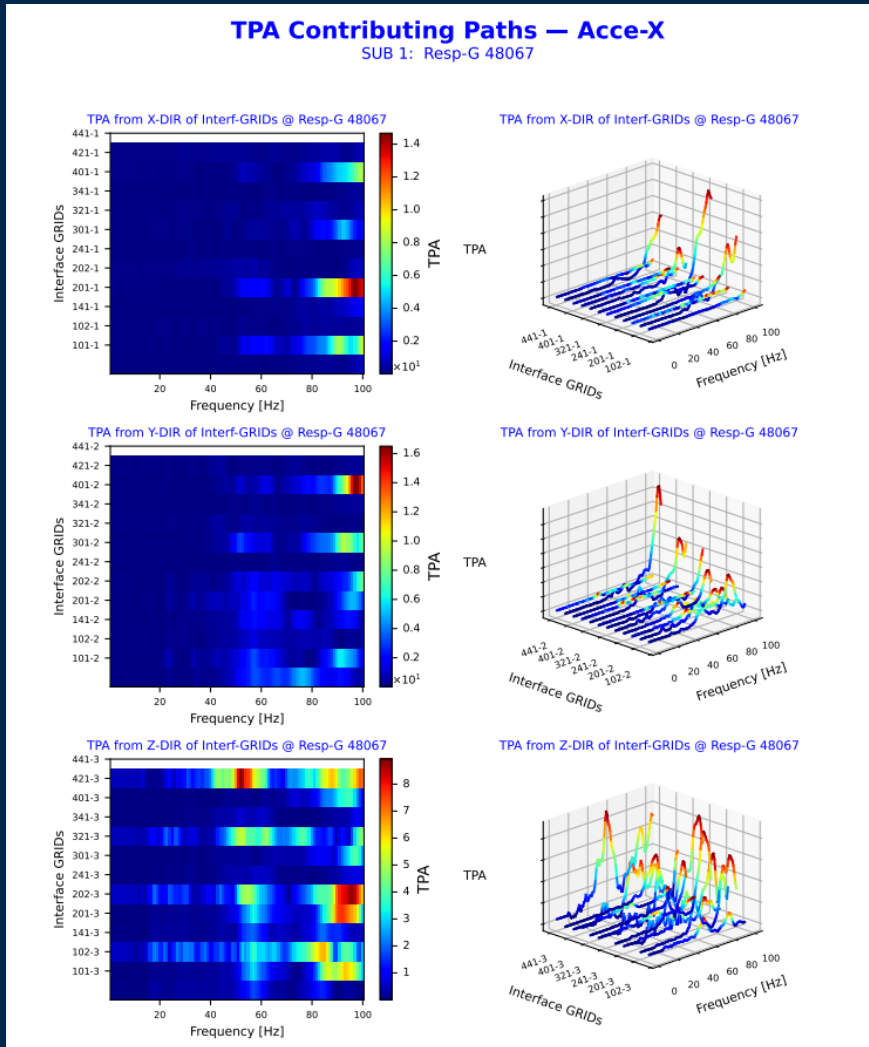
- 935,089 CQUAD4
- 58,017 CTRIA3
- 22,301 CHEXA+CPENTA
- 1,065,542 GRID points

Number of Interface Grids: 12

```
TPADEFN 100      2000      24
+          A      2001
SET3,2000,grid,101,102,141,201,202,241
,301,321,341,401,421,441
SET3,2001,grid,48067
```



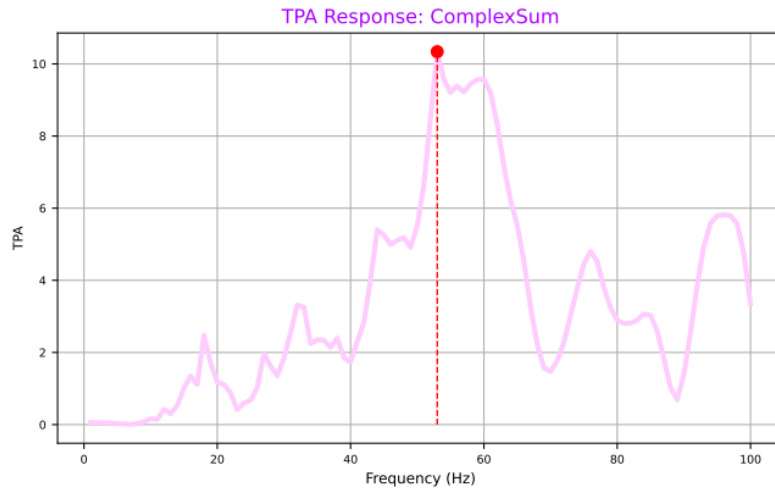
Single-Step TPA results in plots



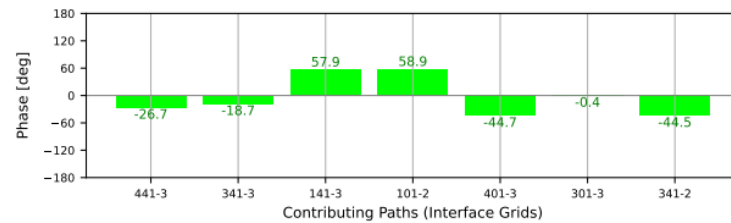
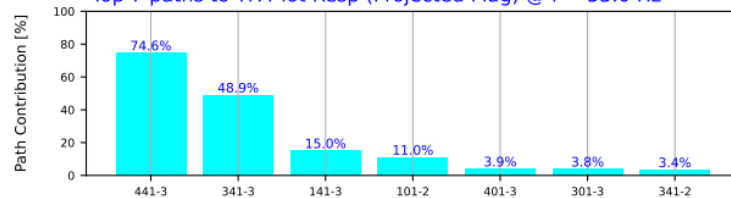
Single-Step TPA results in plots

TPA Tot Resp Contributions: Peak 1 (53.0 Hz) — Acce-X

SUB 1: Resp-G 48067

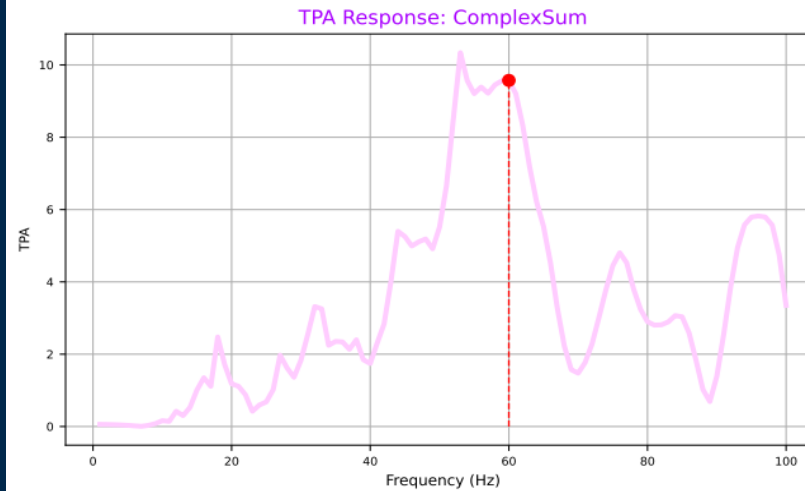


Top 7 paths to TPA Tot Resp (Projected Mag) @ f = 53.0 Hz

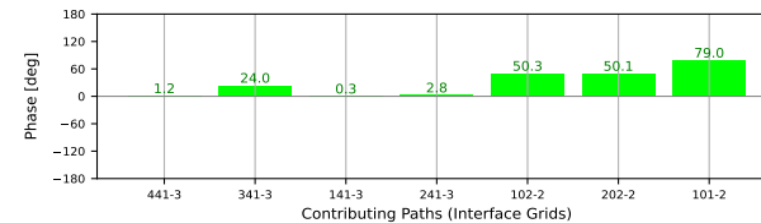
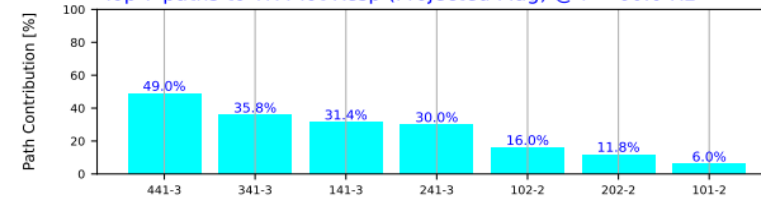


TPA Tot Resp Contributions: Peak 2 (60.0 Hz) — Acce-X

SUB 1: Resp-G 48067



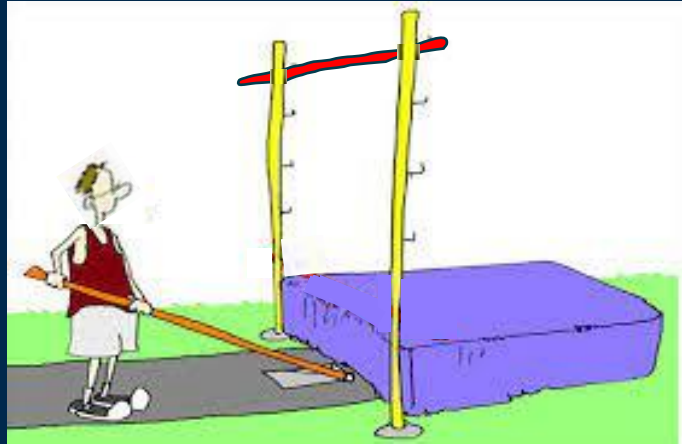
Top 7 paths to TPA Tot Resp (Projected Mag) @ f = 60.0 Hz



Global-Local Analysis using MSC Nastran CARVE

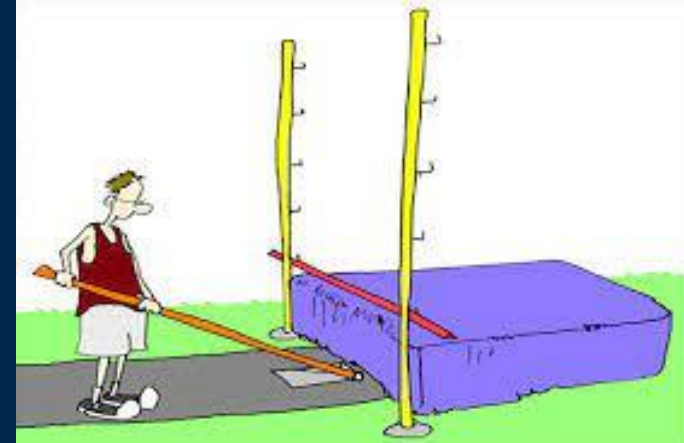
Why Global-Local analysis?

The bar is set quite high for using MSC Nastran superlements



Traditional Superelement analysis

- Define Superelement region
- Define ASET boundary grid points
- SPOINT and QSET
- Assign OP2, OP4, MATDB
- Assign INPUTT2, INPUTT4
- Specify K2GG, M2GG, P2G
- Assemble .asm and .pch files
- Assemble and manage thousands of load cases

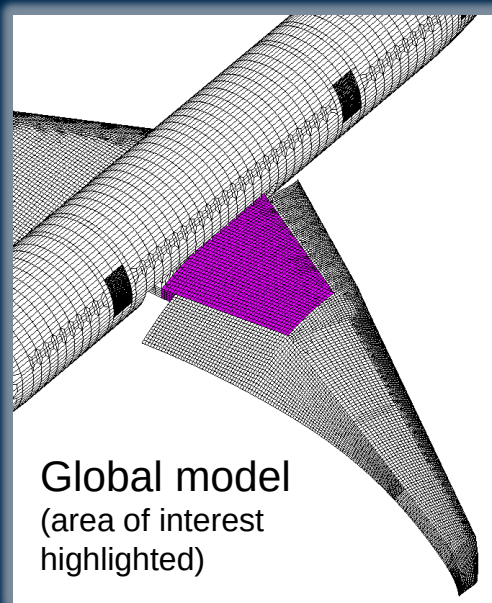


Global-Local Analysis

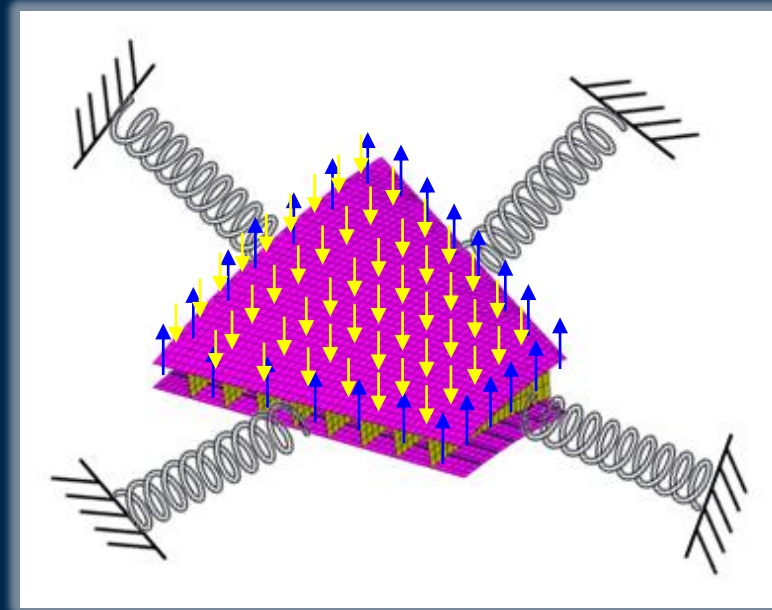
- Automatic generation of boundary stiffness

MSC Nastran CARVE or Global-Local Analysis

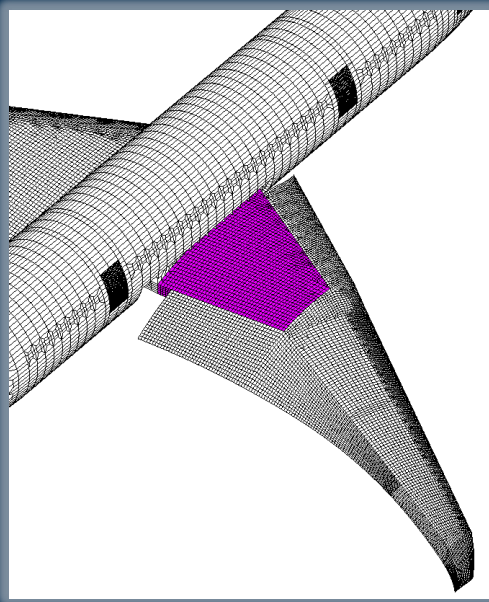
- What it is not
 - Solve the global model and then map displacements from the global model to the boundary of a refined local model
 - You can do this with either Patran or MSC Apex
- What it is
 - “Carving” out a local model with boundary stiffness and boundary loads from the global model



MSC Nastran
CARVE



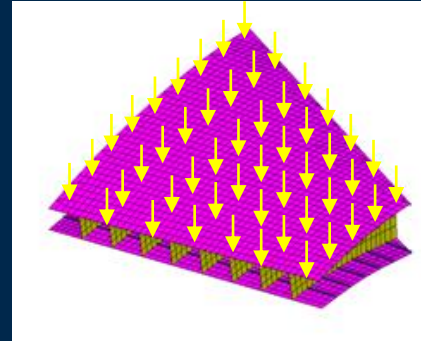
Automated Process



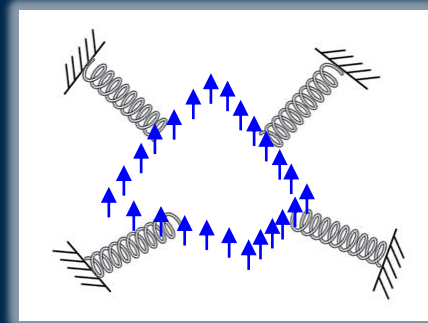
Desire a local representation of the highlighted region and retain the remainder as a DMIG stiffness representation or Super Element



MSC Nastran
CARVE



Local model with local loads

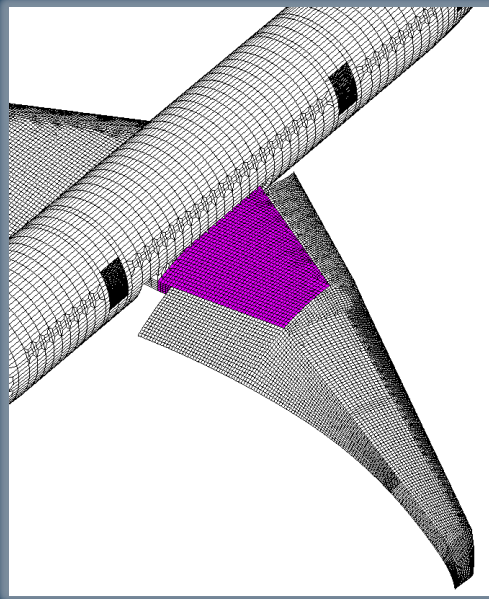


SE with boundary stiffness and boundary loads

MSC Nastran CARVE partitions the model into:

- A local model with local loads
- A Super Element reflecting the boundary stiffness and boundary loads for the remainder of the global model
 - The boundary stiffness can be either represented as a simple DMIG or a full Super Element
 - The boundary loads can be represented as the MSC Nastran loads matrix (PA) or can be output as FORCE/MOMENT entries with IDs consistent with the interior FORCES

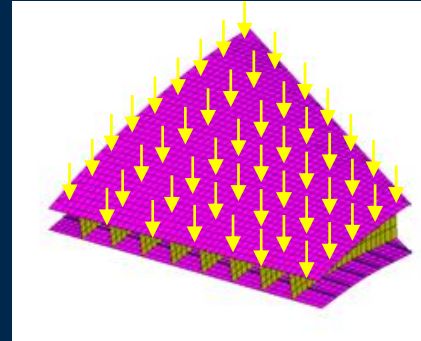
Automated Process



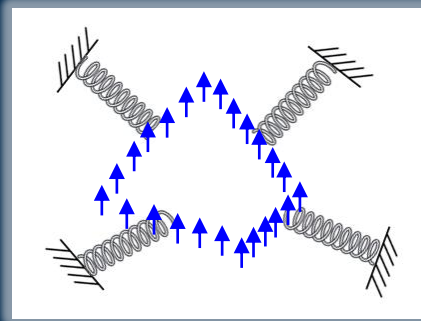
Desire a local representation of the highlighted region and retain the remainder as a super element



MSC Nastran
CARVE

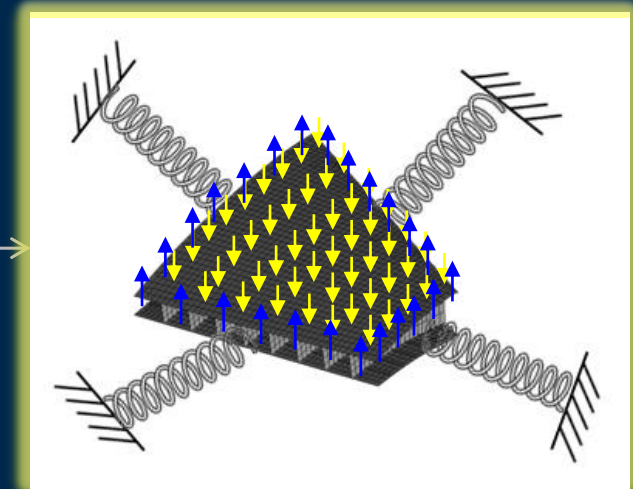


Local model with local loads



SE with boundary stiffness and boundary loads

- A static analysis of the carved local model + SE should yield the same results as the original global model
- Or replace the carved local model with a refined version to investigate additional details in the local area



Refined local model with local loads, boundary stiffness, and boundary loads

What information does MSC Nastran need to perform CARVE?

- Nodes and elements to be included in the local model
 - Defined via **MDSET3** entries
- Name of file to contain the local model
 - This will include the local nodes, elements, local loads, case control, etc., extracted from the global model
 - Defined in FMS section via **ASSIGN CARVE=<my local carve.dat>**
- Boundary stiffness representation, i.e., DMIGPCH or DMIGOP2 or MATDB?
 - Defined via case control **CARVE** statement
- Boundary load representation, i.e., boundary forces & moments or boundary load matrix (PA)?
 - Defined via **PA2FM** selection on the case control CARVE statement
- If using super elements, do you want to enable results recovery in the super element?
 - Parameter **MAKEOTM** set to yes or no

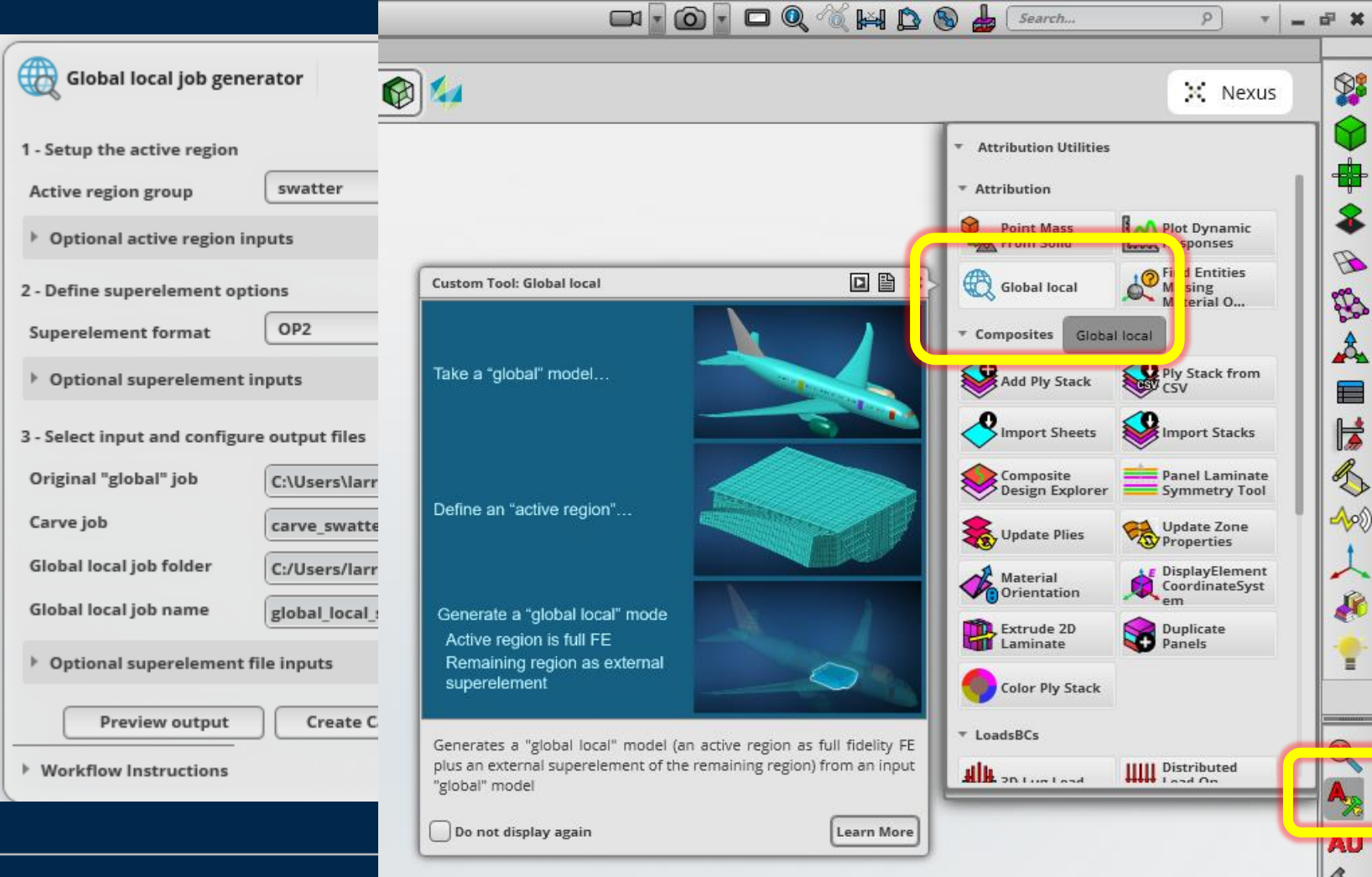


(“there is no maybe, only yes or no” ~Yoda)



MSC Apex Custom Tool to create the CARVE partition job

Easy just got easier ...



to be the local model

the CARVE partition job

data
SC Apex

Real Coupled Modes

Fluid-Structure Analysis in MSC Nastran

MSC Nastran has been able to solve fluid-structure problems for many years.

- "Fluid-structure analysis" can mean:
 - ✓ Vibro-acoustics,
 - ✓ Fluid dynamics (CFD) or explicit cases (slosh, bird strike),
 - ✓ Sometimes, fluid isn't modelled directly (e.g., boundary element methods).

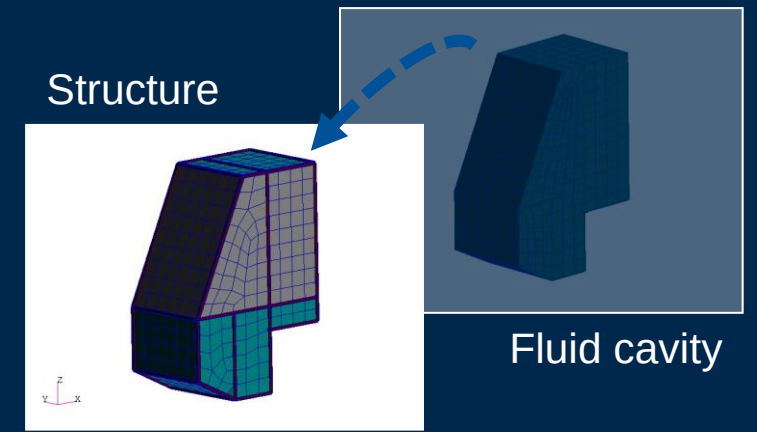
MSC Nastran supports **three types** of fluid-structure analysis:

- ✓ Co-simulation with a separate CFD solver (SCFlow and CoSim).
- ✓ SOL 700 explicit with Dytran (e.g., bird-strike).
- ✓ Coupled/weakly coupled, interior/exterior acoustics (focus here).

Uncoupled Modes

Fluid-structure analysis with light fluids (gases) uses **uncoupled modes**.

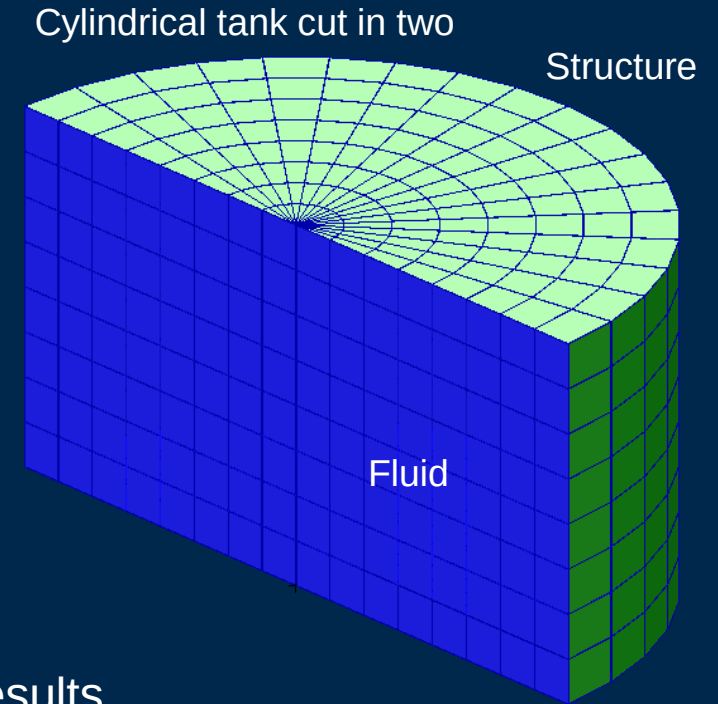
- Fluid modeled with solid elements using MAT10 to set bulk modulus, density, and speed of sound
- Structure modes calculated as if fluid absent (in vacuo).
- Fluid modes calculated as if structure absent
- Modes found with real symmetric eigenvalue solver (e.g., SOL 103)
- For forced response analysis, the physical coupling is modally reduced using the real modes of fluid & structure
- Modal bases computed separately for fluid and structure = **uncoupled modes**



Coupled Modes

Fluid-structure analysis with heavy fluids (liquids) – **coupled modes**

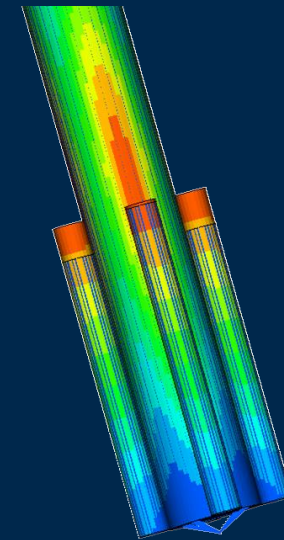
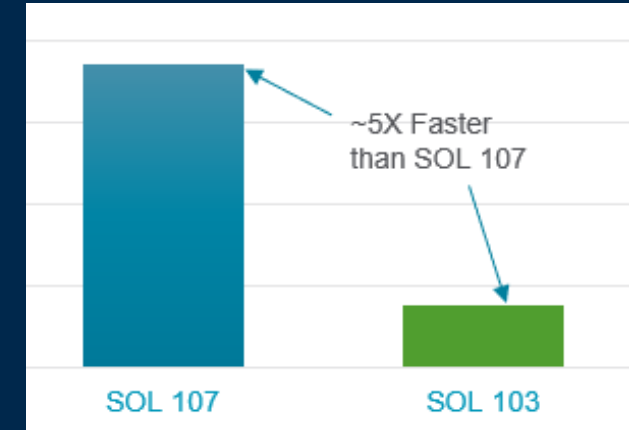
- Heavy fluids (water, gasoline) affect both structure's and fluid's dynamics.
- Heavy fluids stiffen the containing structure and add apparent mass.
- Coupled modes: system modes must be computed together.
- Meshed fluids cause coupling surface, leading to asymmetric matrices.
- Coupled problem: solve asymmetric eigenvalue problem (not with real symmetric solvers).
- Need complex eigenvalue solver (e.g., SOL 107/110).
- Without damping, eigenvalues/vectors are real; solver outputs complex results.
- No reduced response solution using complex modal basis.



Coupled modes in MSC Nastran

Benefits:

- ✓ Real eigenvalue capability for coupled modes
- ✓ Available in SOL 103, SOL 111, and SOL 112
- ✓ No modeling changes needed to use coupled modes
- ✓ Residual vectors can be computed for modal forced response using coupled modal basis (SOL 111/112)
- ✓ Direct force response methods are still available for comparison



Setup and Inputs

For SOL 103, 111, 112, and 200:

For small to medium-sized models, use
METHOD(COUPLED)=

ACMS may be used with this method, and then a
METHOD(STRUCTURE)
and METHOD(FLUID) command are also required

For very large models, use METHOD(SYMCOUPL)=
ACMS is not available with this method

```
SOL 103
METHOD (COUPLED) =10
BEGIN BULK
EIGRL, 10, , , 20
...
ENDDATA
```

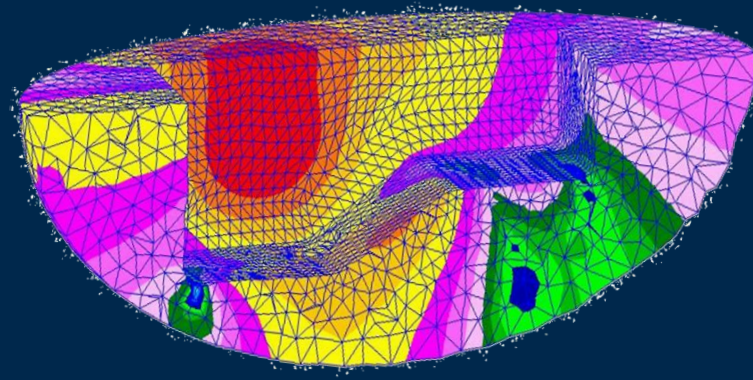
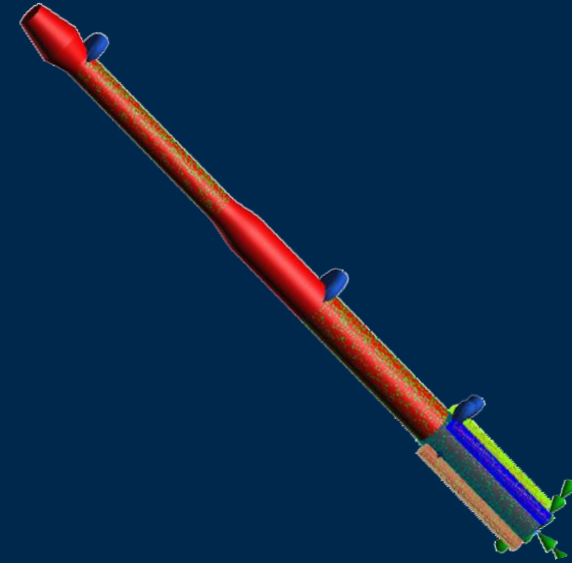
```
DOMAINSOLVER ACMS
SOL 103
METHOD (STRUCTURE) =51
METHOD (FLUID) =52
METHOD (COUPLED) =10
BEGIN BULK
EIGRL, 10, , , 20
EIGRL, 51, , 400.
EIGRL, 52, , 200.

...
ENDDATA
```

```
SOL 103
METHOD (SYMCOUPL) =10
BEGIN BULK
EIGRL, 10, , , 20
...
ENDDATA
```

Typical applications for Coupled Modes

- Liquid-filled reservoirs, for example
 - Hydrazine in rocket launchers and satellite boosters
- Gasoline tank in an automobile
- ICE oil-filled sump
 - Correct vibration modes required for NVH
 - Exterior acoustic field for sound radiation to the far field (e.g., drive past noise)

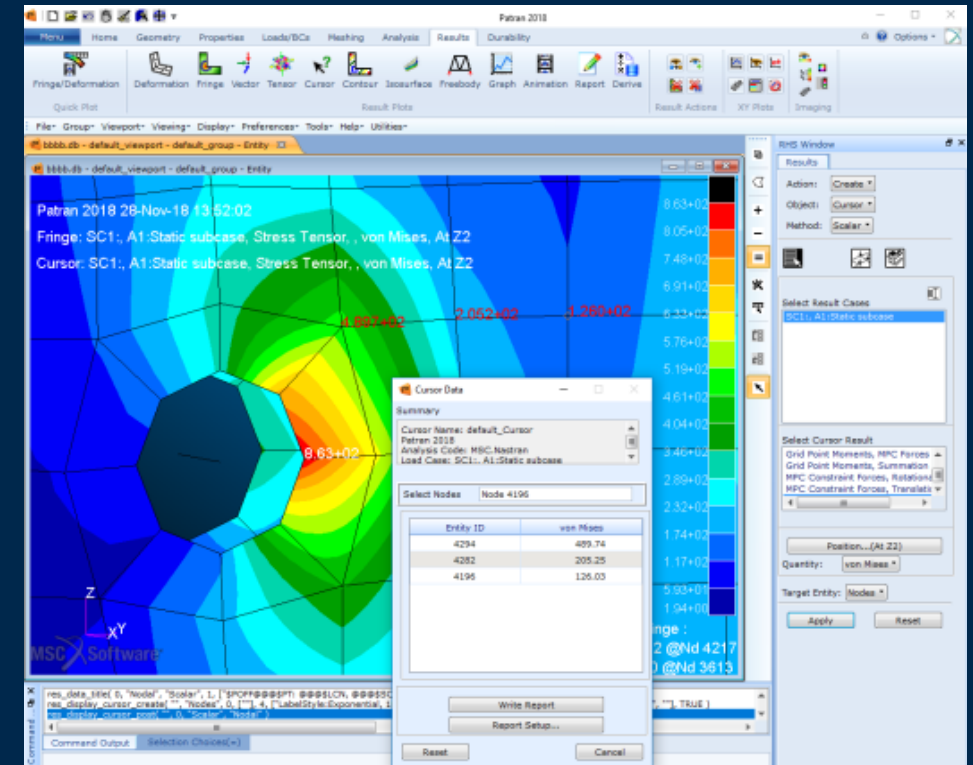


HDF5 Output Modularity

Why you should be using hdf5 output results files

What is HDF5 in MSC Nastran?

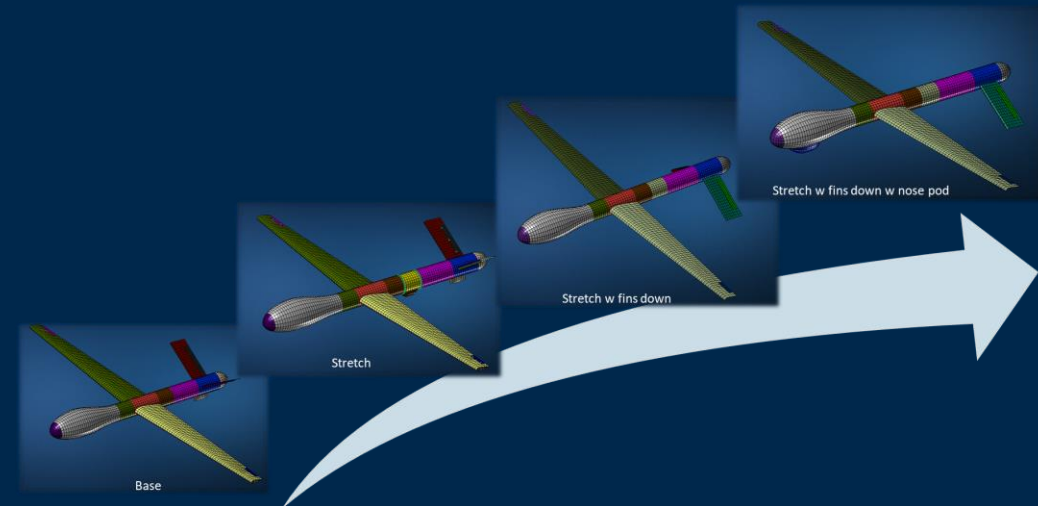
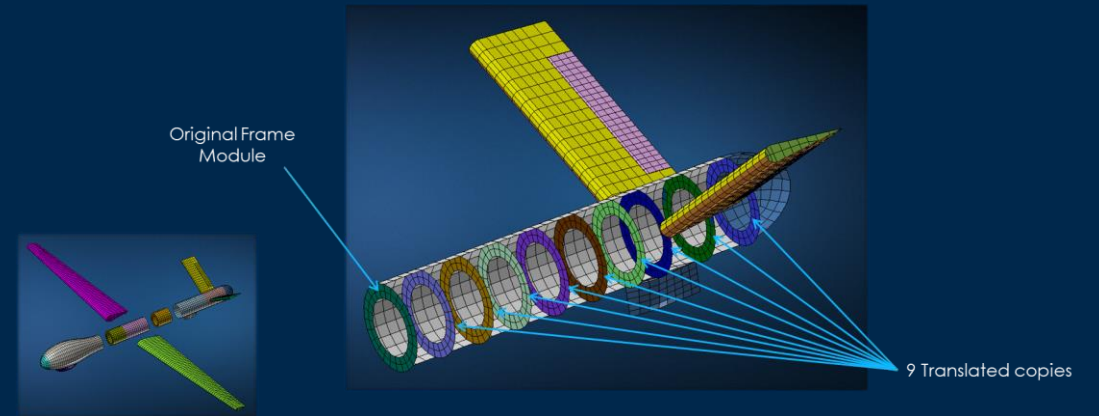
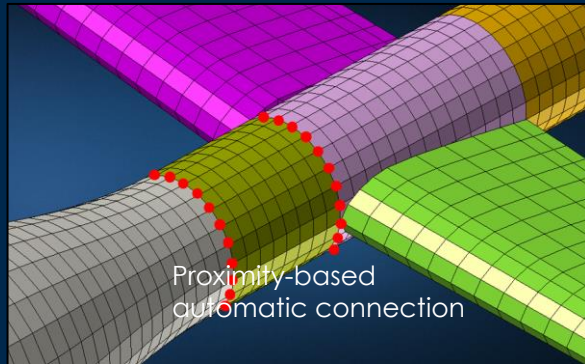
- ✓ The HDF5 file is a container of objects
- ✓ Hierarchical Data Format (HDF5)
- ✓ Open-source, developed by HDF Group
- ✓ Human-readable with HDFViewer and accessible via multiple methods (e.g., scripting)
- ✓ Compressible
- ✓ Control precision of the .HDF5 file (32- or 64-bit) or hybrid (3264)
- ✓ Supported by all major commercial post-processors
- ✓ In MSC Nastran, HDF5OUT, HDF5DA



MSC Nastran Modules

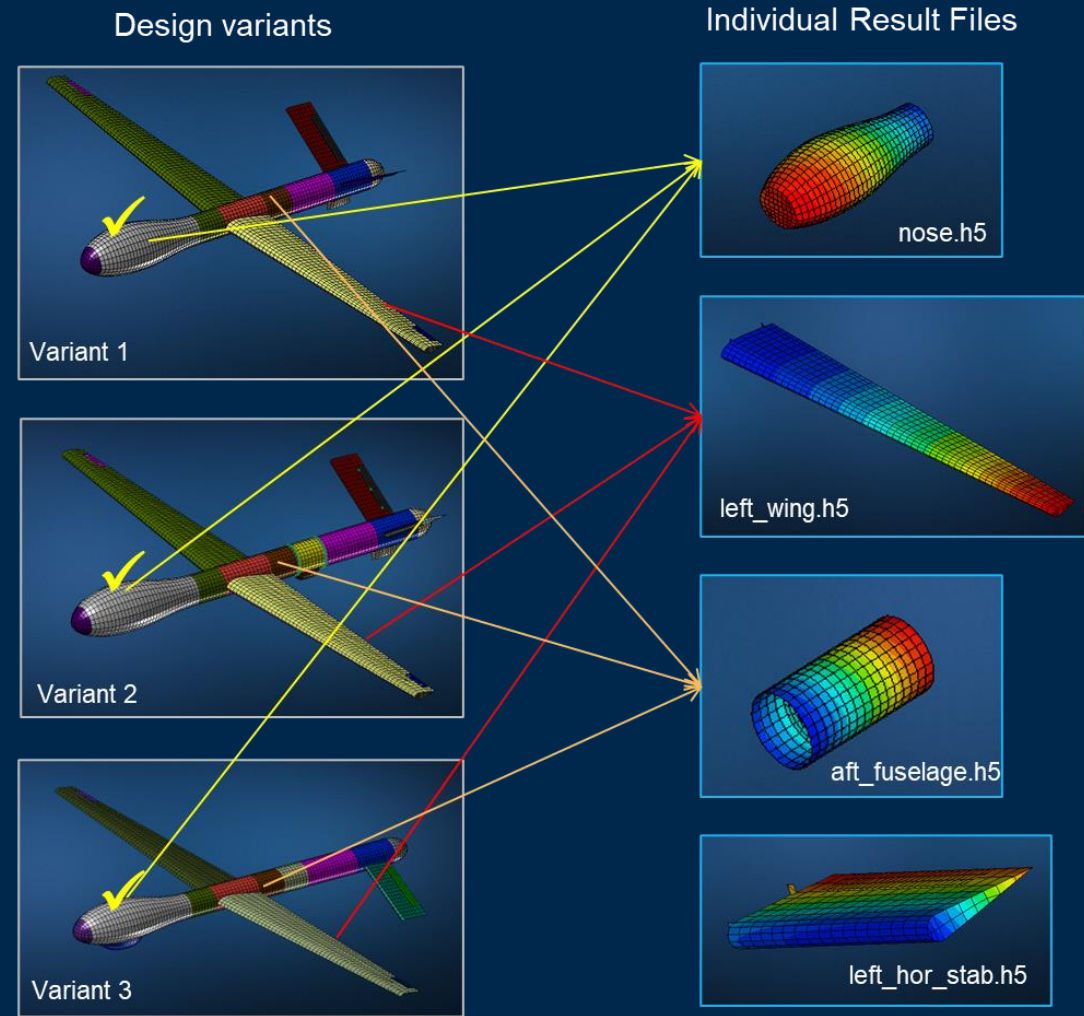
Modules = Parts and Instances for MSC Nastran

- Component models are allowed to have non-unique IDs
- Model a component once and reuse it
- Translation, mirror, rotation, superelement reduction
- Automatic or semi-automatic connections
- Modular post-processing (separate hdf5 file per module or results appending)



Modular results

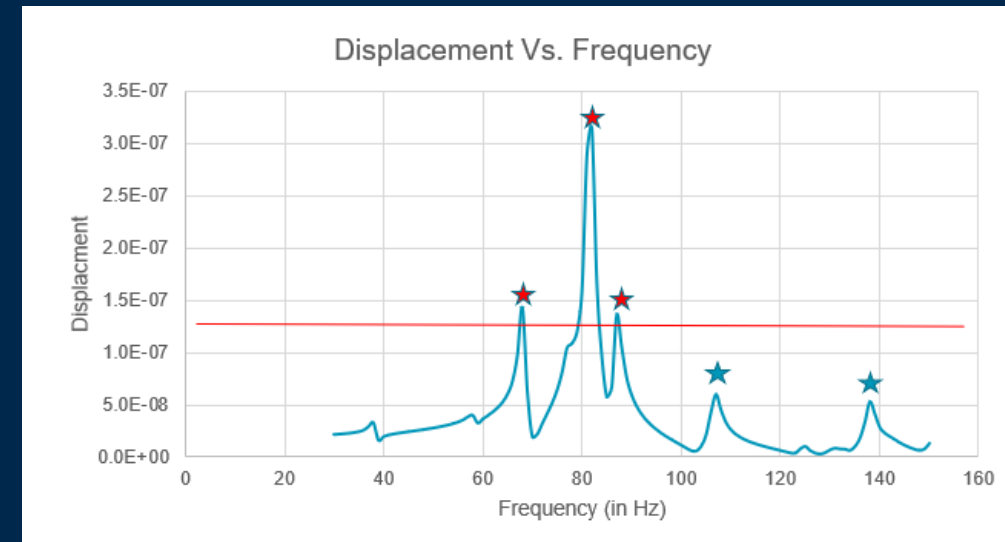
- A component model may appear in many design variants with thousands of load cases
- Module results can be output to a single results file or to individual HDF5 files
- Additional runs can append results to the same HDF5 file for the component
- Component-level result files make it easier to share results data with the component teams
 - Facilitates enveloping and searches within only specific components
 - Faster and less error-prone



ODS and Full Displacement Output using PEAKOUT

ODS and Full Displacement Output using PEAKOUT

- In typical Automotive NVH simulations, users generate Transfer Function results in the frequency domain using SOL 111 with ACMS.
- Output is often at a small set of grid points.
- Sometimes, full model or many grids/elements results are needed at specific frequencies.
- Full model displacements at a frequency are called ODS (Operating Deflection Shape).
- MSC Nastran offers ODS results and full displacement outputs using PEAKOUT with full automation



ODS and Full Displacement Output using PEAKOUT

Generate full model results efficiently using ACMS in SOL 111

Lanczos solver is also supported

Frequencies for results generation can be determined using a SET with specific frequencies or using PEAKOUT for full automation

Variety of results including displacement, velocity, acceleration, stress, strain, element strain energy (ESE)

Both structure-only or fluid-structure models can be used

Usage

There are 3 different use cases:

1. ODS requested for a defined SET of frequencies without the PEAKOUT command

The frequencies to be used are defined in a SET

SET 10=100.,200.,300.

This set may be different in different SUBCASES

2. ODS is requested for PEAKOUT frequencies in combination with PFMODE/GRID/PANEL

The frequencies to be used are detected by PEAKOUT

The FRF used in the peak search are identified by the various PF commands

The frequencies may be different in different SUBCASES

3. ODS is requested for PEAKOUT frequencies without PFMODE/GRID/PANEL

The frequencies to be used are detected by PEAKOUT

The FRF used in the peak search are identified by the new PEAKFRFF/PEAKFRFS option on PEAKOUT

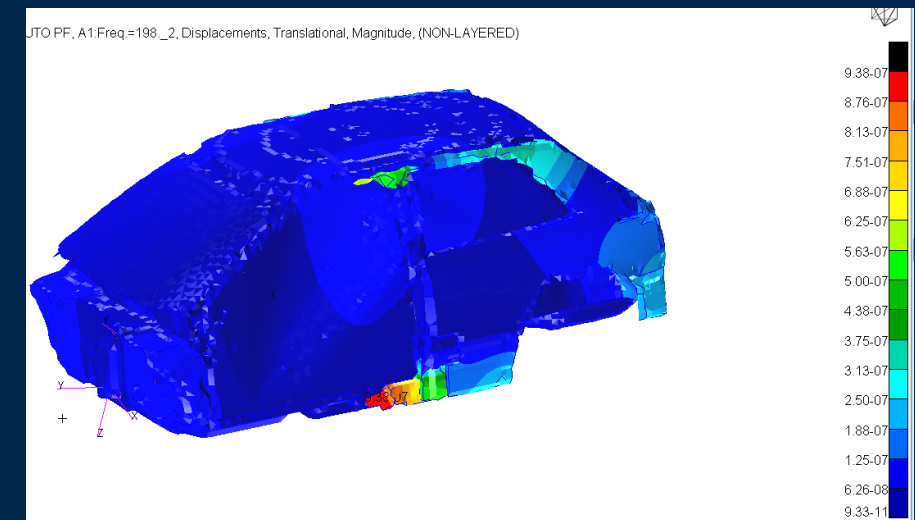
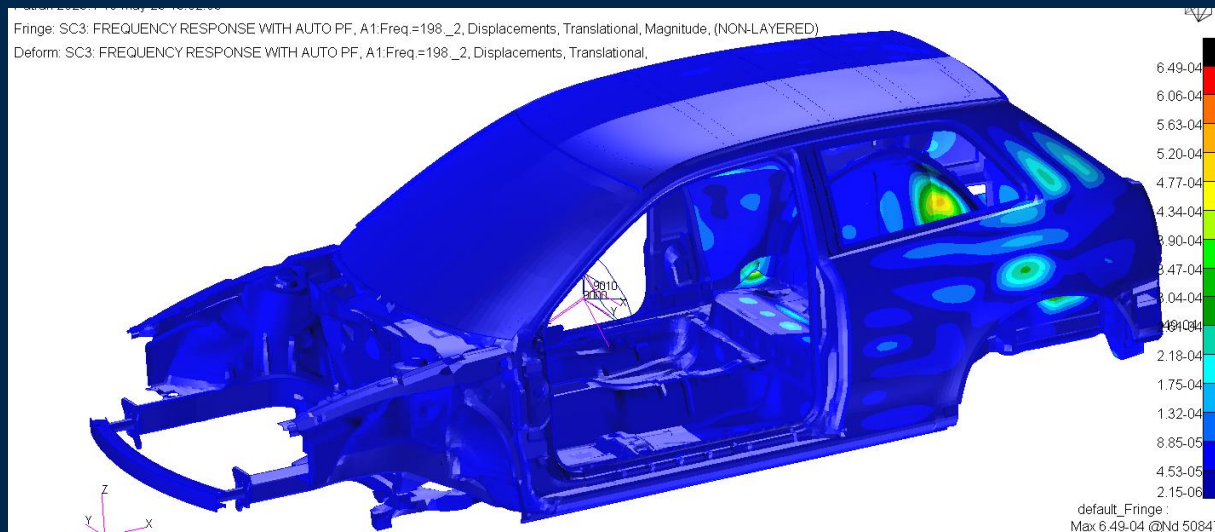
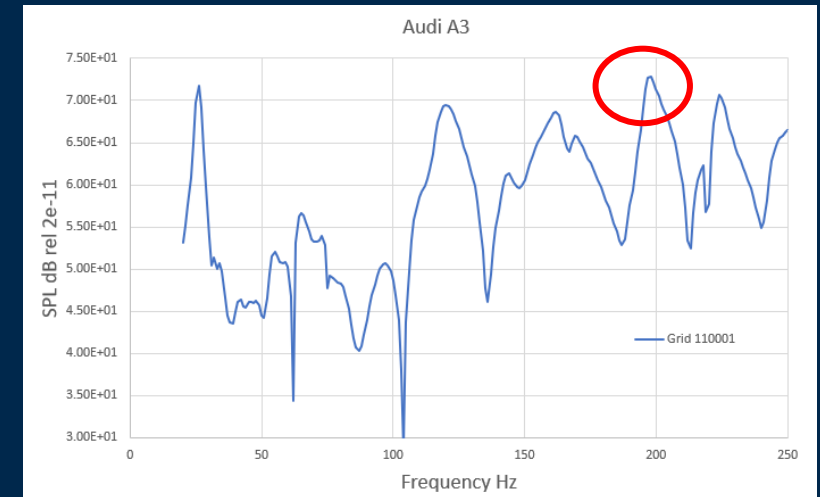
The frequencies may be different in different SUBCASES

Output

- The results may be output to standard Nastran files
 - ✓ PRINT(f06) and PUNCH
 - ✓ PLOT to h5 and op2 (post,-1 or 1)
- In case of op2 output, the ODS results will be written to a separate .op2 file with the name *jobname_ods.op2*
 - ✓ All other results are output to the standard op2 file
 - ✓ **PARAM, PFODSO2,1 will direct also Participation factors to the separate op2 file (not default)**

Example of a car body

- Acoustic analysis of a car body
- PEAKOUT is used to identify the maximum response frequency
- ODS Displacements calculated at that frequency (198Hz)
- Structure Displacements and fluid pressure post-processed in Patran

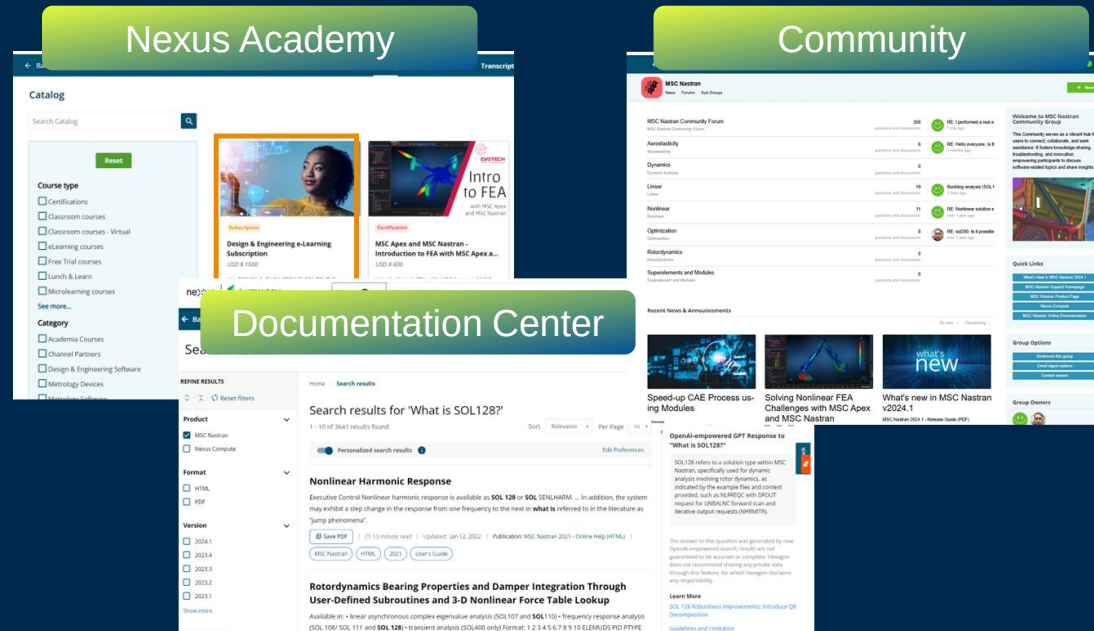
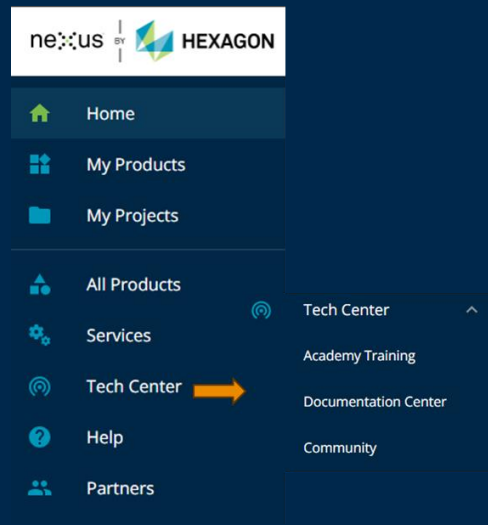


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Nexus

Nexus Home Page (<https://nexus.hexagon.com/>)

- **Free to register**
- MSC eLearning account migrated to **Nexus Academy** <https://nexus.hexagon.com/home/hmi-learning-center-transition/>
- SimCompanion Forums migrated to **Nexus Community** <https://nexus.hexagon.com/home/simcompanion-forums-transition/>



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SimCompanion, Hexagon.com



Support Email:
mscnastran.support@mscsoftware.com

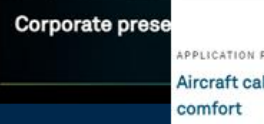


Licensing Questions:
msc_lic.support@mscsoftware.com

Resource library

Access case studies, white paper, magazines, podcasts, videos and more.

MSC Nastran Resource Library

 ARTICLE Actran helps to predict the vibro-acoustic response...	 APPLICATION REPORT Aero engine blade thermo-structural design	 APPLICATION REPORT Aero engine multibody dynamics simulation
 Corporate presentation Aircraft cabin noise passenger comfort	 CASE STUDY Airframe Designs, Ltd	 CASE STUDY Alson E. Hatheway Inc (AEH): Nastran
 CASE STUDY Avio ensure the VEGA launcher's structural integrity	 OTHER Battery Modelling and Design Solutions	 CASE STUDY Boiler Structure: Static Simulation...

Thank you

Name

- Vicky Tsianika, PhD, Multiphysics
Product Group Lead
- Vicky.tsianika@hexagon.com



MSC Nastran TRIVIA

How well do you know your MSC Nastran?



1. In which Case Control entry can I find the Automatic RBE leg trimming functionality?

GEOMCHECK

BCONCHK

FEMCHECK

GROUNDCHECK

Did you know that



Automatically trimming RBE3 legs with the FEMCHECK parameter can drastically speed up your MSC Nastran simulations? When you have large RBE3 elements with many connected grids, this feature automatically removes unnecessary RBE3 legs - reducing runtimes significantly, while still giving you reliable and accurate results. See FEMCHECK RBE3TLVL RBE3THR for details.

2. Which Case Control entry can I use to automatically skin the outer surfaces of solid elements?

SKINELM

SKINSLD

SLDSKIN

SKINPCH

Did you know that



MSC Nastran now lets you automatically skin the free faces of solid elements using the SKINSLD command? Not only does this eliminate manual pre-processing, but it also creates smaller, focused output files and allows for much faster data recovery, since you're only working with the results you really need. Plus, skinned results can be output into a separate .bdf file for reuse in future simulations.

3. True or False. The ADAMSMNF file includes fluid mass, stiffness, damping and fluid-structure interface matrices.

True

False

Did you know that



MSC Nastran supports the direct recovery of acoustic pressure in Adams without the need for restart runs. The MNF file contains the uncoupled (and coupled depending on the workflow) fluid-structure modes. After running an Adams transient analysis, you can use the Adams post-processor to recover and plot acoustic pressure at any locations of interest.